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Prof. Roberto Cerbino
Soft Matter Experiments (SoMeX) Lab
Faculty of Physics
University of Vienna
roberto.cerbino@univie.ac.at
Job-ID: 5570

Re: University Assistant (predoctoral) — Soft Matter / Biophysics (Job-ID 5570)

Dear Prof. Cerbino,

I am writing to apply for the predoctoral position in the SoMeX Lab (Job-ID 5570). My background is computational biology and first-principles physical modelling rather than experimental soft matter physics, and I want to be direct about what that means for this application: I have developed validated frameworks for fluid dynamics, optical physics, and biological collective motion that sit inside the lab's core research areas — rheology and microscopy-based dynamics — but I have not operated a rheometer or built a DDM acquisition pipeline. The case I want to make is that the theoretical and computational work I bring is a genuine complement to the experimental programme, not a substitute for it.

First-principles fluid dynamics and rheology. The SoMeX Lab's primary experimental method for characterising soft matter is rheology: measuring how yield stress fluids, colloidal suspensions, and biological gels flow and deform under applied stress. The Gas Giant framework I developed derives fluid dynamics — kinetic theory, Navier-Stokes, viscosity, and diffusion — from a single axiom (bounded phase space geometry) via Kirchhoff circuit laws, with zero empirical parameters. The key result for rheology: viscosity is $\mu = \tau_c \times g$ (partition lag τ_c times coupling strength g), where both quantities are derived from the spectral structure of the system's phase space rather than fitted to experimental data. Validated: viscosity 2.9% MAE across 12 pure liquids; Poiseuille flow 0.9% mean error; adiabatic expansion 0.1–0.3% error; all at zero empirical parameters. The framework also provides a first-principles account of the yielding transition: in a yield stress fluid, the yield stress σ_y corresponds to the partition floor at which the ordered phase of the bounded oscillatory system breaks down irreversibly — a phase transition in the spectral structure, not an empirical threshold. For the lab's programme on yield stress fluids and colloidal glasses, this gives a predictive framework for rheological properties from molecular structure that complements rheometric measurements: the model predicts what the rheometer should measure before the experiment is run, and the measurement validates or corrects the theoretical partition structure.

Optical physics and DDM analysis. Differential Dynamic Microscopy extracts dynamic information from microscopy movies by computing the image structure function $D(\mathbf{q}, \Delta t)$ across a range of wave vectors and lag times. The physical content of D depends on the optical transfer function of the microscope and the light-matter interaction in the sample. The Lagrangian framework proves Beer-Lambert absorption, Kirchhoff circuit laws, and thermal emission as formally identical operations (Resonant Stack theorem; transmittance $T = 1/e = 0.3679$ in all representations). This provides a first-principles model of the optical signal in a DDM experiment: the image contrast that encodes particle dynamics is governed by the same partition geometry that governs bulk absorption. At a practical level, the Helicopter platform implements multi-modal microscopy analysis combining fluorescence, brightfield, phase-contrast, and spectral channels through sequential constraint exclusion — a framework directly applicable to building analysis pipelines for the lab’s spinning-disk and wide-field DDM acquisition systems. I write instrument control and image analysis code in Python; the lab’s open-source fastDDM code base is the kind of project I would contribute to and extend.

Active matter and biological collective dynamics. The SoMeX Lab’s work on cellular dynamics and bacterial collectives addresses the same physical problem as the Olduvai framework: how do collective biological motion patterns arise from the local coupling rules of individual constituents? Olduvai’s axiom is $\oint \mathbf{J} \cdot d\mathbf{A} = 0$ — zero net current through any closed surface in the biological motion field. From this single conservation law, the framework derives: the decomposition of biological motion into supraspinal and peripheral components (validated against 86 nights of wearable sensor data, AUC = 1.00 for Parkinson’s / ataxia / healthy classification); charge conservation rates $Q_{\text{motor}} = 290.9 \text{ mC/s}$ and $Q_{\text{thought}} = 133.5 \text{ mC/s}$ from first principles. The same conservation law structure applies to bacterial collective motion: the current field $\mathbf{J}$ is the velocity field of the bacterial colony, and the constraint $\oint \mathbf{J} \cdot d\mathbf{A} = 0$ is the topological condition for closed-circuit collective motion patterns. This provides a theoretical framework for connecting DDM-measured intermediate scattering functions $f(\mathbf{q}, \Delta t)$ in bacterial suspensions to the underlying conservation laws of the collective dynamics.

Programming and data analysis. I work primarily in Python, with Rust for performance-critical components. My framework implementations include instrument-level interfacing, large-scale data pipeline construction (Ray distributed, >100 GB datasets), GPU-accelerated computation (WebGPU, compute shaders), and image analysis (multi-modal microscopy fusion, CNN-based feature extraction). MATLAB I have used less extensively but can learn quickly in the context of DDM analysis.

Background and research proposal. I hold an MSc in Computational Biology (Universität Tübingen) and an MSc in Pharmaceutical Biotechnology (HAW Hamburg), and I completed doctoral research in Computational Lipidomics at TUM. My degree titles are not Physics, but the physical content of the first-principles frameworks I have built — bounded phase space geometry, Kirchhoff circuit laws, Banach contraction mappings on thermodynamic manifolds — is the core theoretical language of soft matter physics applied in a different domain context. I have attached a two-page research proposal outlining a programme of work connecting the Gas Giant viscosity framework and the Lagrangian optical physics model to DDM-measured rheological dynamics in model soft matter systems (Carbopol yield stress gels, colloidal hard-sphere suspensions). I hold two academic referees available on request. I am legally resident in Germany with full work authorisation and can relocate to Vienna for the June 2026 start date. English is native.

Yours sincerely,

Kundai Farai Sachikonye